MNS Department
QUIZ
Semester: Spring-2024; Section -XX
Course ID: PHY 111; Course Title: Principles of Physics I
Total Marks: 15

Name:	***ID:***

1. Calculate the resultant vector of the three vectors shown in fig. 1. 4 marks
2. Find the angle between $\vec{B}$ (fig. 1) and a vector with co-ordinate (3, -2, -5) 4 marks
3. The coordinate (meters) of the rabbit's position as functions of time t (seconds) is given by $x = -0.31t^2 + 7.2t + 28$. Find the average acceleration at time $t = 10s \; to \; t = 15s$. 4 marks

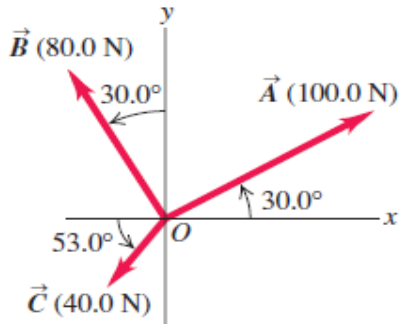

Fig. 1

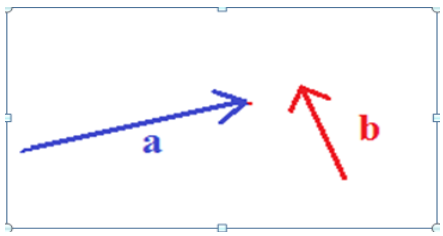

Fig. 2

4. (a) From fig.2 draw the vectors $\vec{a} - \vec{b}$ and $\vec{a} + \vec{b}$ 2 marks

(b) Here are four descriptions of the position (in meters) of a puck as it moves in an *xy* plane:

(1) $x = -3t^2 + 4t - 2$ and $y = 6t^2 - 4t$ (3) $\vec{r} = 2t^2\hat{i} - (4t + 3)\hat{j}$

(2) $x = -3t^3 - 4t$ and $y = -5t^2 + 6$ (4) $\vec{r} = (4t^3 - 2t)\hat{i} + 3\hat{j}$

Which puck (pucks) is (are) moving at constant acceleration? 1 mark